Assignment 3

Problem 1- VAE Synthetic Data with Low-Data Stabilization

Using only **350 real examples per digit**, train a **conditional VAE**.

Before training the VAE:

1. Augment the 350 real examples per digit by **10–20 times** using rotations, shifts, small scaling, and light noise.
2. Train the conditional VAE using the **real + augmented data**.
3. For each digit, sample latent vectors: $z \sim N(0,I)$
4. Generate **5× (1000 examples in each run)** of synthetic samples per digit. Each run should start with different random values of the weights, that should give better variants of generated samples.
5. Pass generated samples through the LeNet-5 (**trained on only the same allowed 350 examples**) classifier and calculate:

   confidence=max (softmax output of the trained LeNet above)
6. Create three synthetic datasets:
   - **Set A:** all generated samples
   - **Set B:** high-confidence samples: confidence ≥ 0.9
   - **Set C:** mid-confidence samples: 0.6 ≤ confidence ≤ 0.9
7. Train LeNet-5 using:
   - 350 real examples per digit
   - plus, the selected VAE-generated samples
8. Compare Set A, Set B, and Set C with:
   - the 350-real baseline
   - the 1000-real baseline
   - the GAN-based results

Question:

Does VAE-generated data benefit more from high-confidence selection or mid-confidence/diverse latent samples, and how does it compare with GAN-generated data and simple augmentation?

Problem 2 - GAN Synthetic Data with Low-Data Stabilization

Using only **350 real examples per digit**, train a **conditional GAN / cDCGAN**.

Before training the GAN:

1. Augment the 350 real examples per digit by **10–20 times** using rotations, shifts, small scaling, and light noise.
2. Train the cDCGAN using the **real + augmented data**.
3. For each digit, generate **5× (each 1000 examples)** of synthetic samples.
4. Pass generated samples through the LeNet-5 classifier and calculate:

   confidence=max (softmax output)
5. Create three synthetic datasets:
   - **Set A:** all generated samples
   - **Set B:** high-confidence samples: confidence ≥ 0.9
   - **Set C:** mid-confidence samples: 0.6 ≤ confidence ≤ 0.9
6. Train LeNet-5 using:
   - 350 real examples per digit
   - plus the selected GAN-generated samples
7. Compare Set A, Set B, and Set C with:
   - the 350-real baseline
   - the 1000-real baseline

Question:

Which GAN selection strategy gives the best improvement, and does GAN-generated data reduce the need for real data? Compare with adding augmented data as in Assignment 2.

Summarize all the results of problem 1 and problem 2 in this table.

		Confidence Level (CL)		
		all generated data	CL >0.9 only	0.6 < CL >0.9
No. of examples				
Model used	VAE			
	GAN			

Problem 3: Understanding the Impact of Attention Mechanisms

a. Build a convolutional neural network (CNN) to classify images from the **ReducedMNIST** dataset (ex.: LeNet-5). Then, build a second version of the CNN that includes a **spatial attention mechanism**. Compare the two models in terms of **accuracy** and **training time**. What difference does the attention mechanism make?

b. Revisit the spoken digits task from Assignment 2, where you used spectrogram images to recognize 10 spoken digits. Build two CNN models for this task: one **with an attention mechanism** and one **without** (which you implemented in ass-2). Compare their performance and explain how attention affects the accuracy and training time in this case.

For both parts (a and b), your submission should include:

- A clear and detailed report describing **(put your answers on a table)**:
  - The network architectures you used
  - Your training process and chosen hyperparameters
  - A comparison of performance between models with and without attention
  - Your analysis of how the attention mechanism affected the results
- Insights and observations you gained from your experiments.
- Suggestions for future improvements, such as trying different types of attention or tuning the model further.

Problem 4: General Information Retrieval (including Q/A)

Information Retrieval and Question Answering using Arabic language. This problem introduces classical keyword-based retrieval, semantic search using embeddings, and Retrieval-Augmented Generation (RAG). You will apply these techniques using a single Arabic book (in plain text format) as the reference source.

Tasks

*(**You must use a single open-source small LLM from HuggingFace — e.g., a model in the range of 1B–3B parameters — throughout all tasks**.)*

1. Book Preparation for Search
   - Select a public domain or permitted Arabic book in plain text format.
   - Be sure that the selected book was not used last year.
   - Split the text into short paragraphs (2–4 sentences).
   - Generate embeddings using a multilingual or Arabic sentence embedding model.
   - Index the vectors using a tool such as FAISS.
2. Retrieval System
   - Build an interface that accepts Arabic queries.
   - Implement:
     - (a) Classical search (TF-IDF or BM25)

    - (b) Semantic search (embedding-based)
- Display the top 5 results from both methods side by side for each query.

3. **RAG System**
- Extend the interface to answer Arabic questions using the same selected small LLM.
- For each query:
    - (a) Retrieve relevant text using semantic search
    - (b) Generate an answer using RAG (LLM + retrieved content)
    - (c) Generate an answer using the LLM only (without retrieval)
- Display both answers for comparison.

Deliverables

1. Description of the selected book:
    - Title and source
    - Text preprocessing and chunking method
    - Embedding and indexing process
2. Results from at least 10 Arabic queries using:
    - Classical search
    - Semantic search
3. Answers for 10 queries using:
    - RAG
    - LLM-only
4. Comparison and reflection:
    - Which retrieval method produced more relevant results
    - Whether retrieval improved LLM-generated answers
    - Conclusions on the value of semantic retrieval and RAG for Arabic text

Important Note

- The model should be downloadable from HuggingFace and runnable locally (CPU or limited resources).
- Choosing a less common book topic is encouraged, as it better highlights the benefit of retrieval.
- Create different kind of questions including direct and indirect, easy and hard ones to get good insights.

Problem 5: Benchmarking Arabic NLP Tasks Across LLMs

Learning Objective

The goal of this assignment is to evaluate how different Large Language Models (LLMs) perform on Arabic NLP tasks, to identify their strengths and weaknesses, and to understand the specific challenges of processing Arabic, including dialect variation, morphology, and ambiguity.

Models to be Evaluated: ChatGPT, Gemini, ALLaM, Jais, Fanar

Task Assignment: Each group is assigned one unique task from the list of 27 Arabic NLP tasks.

1. Scope Definition: Each group must clearly define the task objective (such as accuracy, robustness, or hallucination), specify the application domain, and describe the data sources used.

The dataset must include a balanced mix of Arabic varieties:

- approximately one third of Modern Standard Arabic
- one third Classical Arabic
- one third Arabic dialects

2. Dataset Creation: Each group must construct a dataset according to the predefined number of cases for the assigned task.

Data requirements:

- Real data should form the **majority of the dataset**

Each sample must include a correct reference (gold) output.

Annotation:

- At least two annotators per sample
- Disagreements resolved by majority voting (a 3rd annotator when needed)
- Report on a simple inter-annotator agreement (IAA) measure

The dataset must be clean, balanced, and well documented.

3. Evaluation Protocol: All five models must be evaluated using the same inputs and the same prompt format. Model settings should be kept as consistent as possible.

Each group must record and submit inputs, prompts, outputs

4. Evaluation Metrics: Each group must use the predefined metric for its task and report both quantitative results and brief qualitative observations.

5. Error Analysis: Each group must analyze model failures by including:

- at least three incorrect or weak outputs
- at least three Arabic-specific challenges (e.g., dialect confusion, morphology, ambiguity)
- at least two difficult or borderline cases

Each case must be explained clearly.

6. Results Presentation

Each group must present:

- a comparison table across the five models
- at least one visualization (such as a bar chart)
- Selected output examples

7. Report Requirements

Each group must submit a report of no less than eight pages including:

- task definition
- dataset creation
- annotation process
- evaluation setup
- results
- error analysis
- limitations
- final conclusions, comments and recommendations.

Important Rules

- Each group must work on a different task and must evaluate all five models. All datasets must be original, and full logs must be submitted.
- Below is a guideline checklist to be sure that the Benchmarking process is reasonably done. Each team should include in the report what they could apply of this checklist.

#	اسم المهمة بالعربية	Task in English	الوصف بالعربية	عدد حالات الاختبار	المقياس + الرابط	مرجع (Wikipedia)
1	سؤال وجواب	Question Answering	الإجابة عن أسئلة اعتمادًا على نص أو معرفة عامة والمطلوب التأكد من دقة الإجابة	100 سؤال + إجابة مرجعية	F1 / Exact Match	https://en.wikipedia.org/wiki/Question_answering
2	التلخيص	Summarization	تلخيص نص عربي مع الحفاظ على المعنى	30 فقرة لا تقل الواحد عن 50 كلمة، لكل فقرة ملخص مرجعي	ROUGE	https://en.wikipedia.org/wiki/Automatic_summarization
3	ترجمة إنجليزي-عربي	English-to-Arabic Translation	ترجمة جمل إنجليزية إلى العربية	50 جملة + ترجمة مرجعية	BLEU	https://en.wikipedia.org/wiki/Machine_translation
4	التشكيل	Diacritization	إضافة الحركات	50 جملة من 10–20 كلمة	CER / Accuracy	https://en.wikipedia.org/wiki/Arabic_diacritics
5	تحليل المشاعر	Sentiment Analysis	تصنيف النص إلى إيجابي/سلبي/محايد	100 جملة قصيرة	Accuracy / F1	https://en.wikipedia.org/wiki/Sentiment_analysis
6	كشف العواطف	Emotion Detection	تحديد العاطفة: فرح، غضب، حزن، خوف...	100 جملة	F1	https://en.wikipedia.org/wiki/Emotion_classification

7	تصنيف الموضوع	Topic Classification	تصنيف النص حسب المجال	80 نصًا قصيرًا	Accuracy	https://en.wikipedia.org/wiki/Text_classification
8	الإعراب	parsing	الإعراب لكلمات النص	50 جملة من 5-15 كلمة في كل جملة	word accuracy, human evaluation	check internet
9	التعرف على الكيانات	Named Entity Recognition	استخراج الأشخاص والأماكن والمؤسسات	80 فقرة	F1	https://en.wikipedia.org/wiki/Named-entity_recognition
10	استخراج الكلمات المفتاحية	Keyphrase Extraction	استخراج أهم الكلمات أو العبارات	80 فقرة	Precision@k	https://en.wikipedia.org/wiki/Keyword_extraction
11	التشابه النصي	Text Similarity	تقدير التشابه بين نصين	80 زوج جمل	Pearson	https://en.wikipedia.org/wiki/Semantic_similarity
12	كشف إعادة الصياغة	Paraphrase Detection	هل الجملتان تحملان نفس المعنى؟	80 زوج جمل	Accuracy / F1	https://en.wikipedia.org/wiki/Paraphrase
13	توليد نص عربي	Arabic Text Generation	إنتاج نص عربي حول موضوع محدد	30 prompt (فقرات تعليمية لتوضيح مبدأ محدد)	Human Evaluation	https://en.wikipedia.org/wiki/Natural-language_generation
14	إعادة الصياغة	Paraphrasing	إعادة كتابة الجملة بنفس المعنى	100 جملة	BLEU + تقييم بشري	https://en.wikipedia.org/wiki/Paraphrase
15	تبسيط النص	Text Simplification	تبسيط النص مع حفظ المعنى	40 فقرة	SARI	https://en.wikipedia.org/wiki/Text_simplification
16	توليد أسئلة	Question Generation	توليد أسئلة من فقرة معطاة	40 فقرة، 3 أسئلة لكل فقرة	ROUGE	https://en.wikipedia.org/wiki/Question_generation
17	توليد حوار	Dialogue Generation	إنتاج حوار مناسب لسياق معين	20 سيناريو (كل حوار من 3-5 دورات سؤال وجواب في نفس الموضوع)	Human Evaluation	https://en.wikipedia.org/wiki/Dialogue_system
18	الوسوم النحوية	POS Tagging	تحديد نوع كل كلمة	100 جملة	Accuracy	https://en.wikipedia.org/wiki/Part-of-speech_tagging
19	التحليل الصرفي	Morphological Analysis	تحليل بنية الكلمة العربية	100 كلمة	Accuracy	https://en.wikipedia.org/wiki/Morphological_analysis
20	الإرجاع للأصل	Lemmatization	إرجاع الكلمة إلى صيغتها الأساسية	100 كلمة	Accuracy	https://en.wikipedia.org/wiki/Lemmatisation
21	استخراج الجذر	Root Extraction	استخراج الجذر العربي للكلمة	100 كلمة	Accuracy	https://en.wikipedia.org/wiki/Arabic_morphology
22	تمييز اللهجات	Dialect Identification	تحديد اللهجة العربية	200 جملة من 5 لهجات عربية على الأقل	Accuracy	https://en.wikipedia.org/wiki/Arabic_dialects
23	اضافة علامات الترقيم	Adding Punctuation marks	اضافتها لنص خام ليس به علامات ترقيم	60 فقرة قصيرة	F1	https://en.wikipedia.org/wiki/Punctuation
24	استخراج المعلومات	Information Extraction	تحويل النص إلى حقول منظمة	60 فقرة قصيرة	F1	https://en.wikipedia.org/wiki/Information_extraction
25	تصنيف الوثائق	Document Classification	تصنيف نص طويل إلى فئة	50 مقالًا (حدد 8 مجالات مختلفة: سياسة - اقتصاد..	Accuracy	https://en.wikipedia.org/wiki/Document_classification
26	التصحيح اللغوي	Grammar & Spelling Correction	تصحيح الأخطاء الإملائية والنحوية	100 جملة بها أخطاء	GLEU	https://en.wikipedia.org/wiki/Grammar_checker
27	كشف الإساءة	Toxicity Detection	كشف النصوص المسيئة أو العدائية	100 جملة أو تعليق	F1	https://en.wikipedia.org/wiki/Toxicity_(behavior)

A Checklist Table

(recommended to be used in your answer)

#	Category	Checklist Item	Status (Yes/No)	Notes
1	Scope Definition	Have you clearly defined the NLT task(s) to be evaluated?		
2	Scope Definition	**Have you specified the evaluation goals (accuracy, robustness, etc.)?**		
3	Language and Domain Selection	Have you selected relevant languages and dialects?		
4	Language and Domain Selection	Have you included multiple genres and domains?		
5	Data Collection	Is the data licensed for research/benchmark use?		
6	Data Collection	Have you included real-world or synthetic data?		
7	Annotation	Is there a clear annotation schema for the task?		
8	Annotation Quality	Is inter-annotator agreement calculated and acceptable?		
9	Data Balance	Does the dataset reflect demographic and topical diversity?		
10	Preprocessing	Is the data cleaned and normalized consistently?		
11	Preprocessing	Is the preprocessing procedure documented?		
12	Evaluation Metrics	Are appropriate metrics defined for each NLT task?		
13	Evaluation Metrics	Do you include baseline results or evaluation scripts?		
14	Documentation	Is the data creation process fully documented?		
15	Documentation	Are known limitations or biases discussed?		
16	Fairness	Does the dataset include diverse and fair representation?		